MCMP117 · Fundamentals of Operating Systems

Final Exam --- Sample Paper

Istanbul Okan University · Vocational School · Electronical Communication Technology Programme

Instructor: Asst. Prof. Dr. Hasan Oğuz Date: [to be announced] Duration: [90] minutes

Name, surname: _____ **Student ID:** _____

Signature: _____ **Seat / room:** _____

Part	A	B	C	D	E	F	**Total**
Points	20	12	36	14	9	9	**100**
Score							

Rules. Closed book [to be confirmed]. [A basic, non-programmable calculator is allowed — to be confirmed.] Phones off. Write your answers in the boxes; show every step of a calculation (formula → numbers → result with unit). Scheduling conventions of the course: context-switch time 0; ties broken by arrival time, then by name; a smaller priority number means a higher priority; in Round Robin a process arriving at the moment a quantum expires enters the queue before the preempted process. TLB: use EAT $= t_{\text{TLB}} + (2-\alpha)\,t_m$. Linux questions refer to the practice folder `~/lab` (created by `mcmp117_lab_setup.sh`); your current directory is `~/lab` unless stated otherwise.

Part A --- Multiple choice (10 × 2 = 20 points)

Circle *one* answer.

A1. UNIX was created at Bell Labs in (a) 1956 (b) 1969 (c) 1981 (d) 1991

A2. In which OS structure does only a minimal kernel run in kernel mode, with most services in user space? (a) monolithic (b) simple (c) microkernel (d) modular

A3. Which of these is *not* stored in a process control block? (a) process state (b) program counter (c) list of open files (d) the source code of the program

A4. The time from a process's arrival until it first gets the CPU is its (a) turnaround time (b) waiting time (c) response time (d) burst time

A5. Which semaphore operation can block the calling thread? (a) `signal(S)` (b) `wait(S)` (c) both (d) neither

A6. The addresses generated by the CPU while a program runs are (a) physical (b) logical (c) disk-block numbers (d) inode numbers

A7. Thrashing means (a) very high CPU utilisation (b) constant page faults and low CPU utilisation (c) a TLB hit ratio of $100\,\%$ (d) a full disk

A8. After `chmod 754 f`, `ls -l f` shows (a) `-rwxr-xr--` (b) `-rwxr-x---` (c) `-rwxrw-r--` (d) `-rw-r-xr--`

A9. `cmd 2> /dev/null` (a) discards the normal output (b) discards the error messages (c) writes errors to a file named 2 (d) appends the output

A10. Which command lists the background jobs of the current shell? (a) `ps` (b) `top` (c) `jobs` (d) `bg`

Part B --- Short answers (3 × 4 = 12 points)

B1. A program calls `write()`. Describe, step by step, how the call reaches the kernel and returns (API/wrapper, system-call number, trap, mode bit, system-call table). **[4 pt]**

B2. What is a race condition? Use `counter++` as the example, and name the three requirements that a solution to the critical-section problem must satisfy. **[4 pt]**

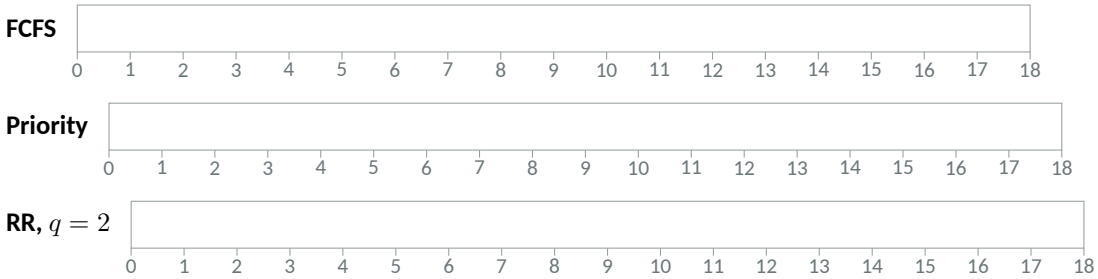
B3. What is thrashing, and how does the working-set model help the OS to prevent it? **[4 pt]**

Part C --- Numerical problems (36 points)

C1. Four processes (times in ms):

Process	P1	P2	P3	P4
Arrival	0	2	3	5
Burst	6	2	7	3
Priority	3	1	4	2

For each algorithm draw the Gantt chart and compute the average waiting time and the average turnaround time: (a) FCFS, (b) preemptive Priority, (c) Round Robin with $q = 2$. **[12 pt]**



C2. Pages are 2 KB. The page table is: page $0 \to$ frame 3, $1 \to 8$, $2 \to 6$, $3 \to 1$ (four entries). (a) Translate the logical address 5000 into a physical address. **[3 pt]**
(b) What happens for the logical address 9000? **[1 pt]**
(c) $t_{\text{TLB}} = 10$ ns, $t_m = 90$ ns, $\alpha = 0.8$: compute the effective access time. **[2 pt]**
(d) Free holes of 200, 450, 100, 300, 500 KB (in this order). Which hole receives a request of 280 KB with first-fit, best-fit and worst-fit? **[2 pt]**

C3. Reference string $5, 4, 1, 5, 0, 5, 4, 0, 2, 1, 0, 1$ with 3 frames. Fill in the frame tables and give the number of page faults for FIFO, OPT and LRU. **[8 pt]**

FIFO

ref	5	4	1	5	0	5	4	0	2	1	0	1
fr. 0												
fr. 1												
fr. 2												
fault												

faults: _____

OPT

ref	5	4	1	5	0	5	4	0	2	1	0	1
fr. 0												
fr. 1												
fr. 2												
fault												

faults: _____

LRU

ref	5	4	1	5	0	5	4	0	2	1	0	1
fr. 0												
fr. 1												
fr. 2												
fault												

faults: _____

C4. (a) $t_m = 100$ ns, $t_{\text{fault}} = 5$ ms, page-fault rate $p = 0.0005$: compute the effective access time. How many times slower than RAM is it? **[3 pt]**

(b) A file has records of 50 bytes and blocks of 512 bytes, both numbered from 0. In which block and at which offset does record 100 start? Does the record fit in that block? **[3 pt]**

(c) Jobs wait for I/O half of the time ($p = 0.5$): compute $U(4)$. A program is $25\,\%$ serial: what speedup does Amdahl's law allow on 3 cores, and what is the limit for many cores? **[2 pt]**

Part D --- Write the command (7 × 2 = 14 points)

Write *one* command line for each task.

D1. Long listing of `docs`, including hidden files, with human-readable sizes.

D2. Rename `docs/notes_01.txt` to `docs/meeting_01.txt`.

D3. Create the empty files `report1.txt` … `report5.txt` with one short command.

D4. Create a symbolic link `latest.log` in the current directory that points to `logs/access.log`.

D5. Add execute permission for the owner (only) of `scripts/backup.sh`, in symbolic mode.

D6. The process with PID 4321 ignores SIGTERM. Stop it anyway.

D7. Show how much disk space the directory `logs` uses, in human-readable form.

Part E --- Write the pipeline (3 × 3 = 9 points)

E1. `data/cities.txt` has one city per line. Print the most frequent city together with its count.

E2. How many requests in `logs/access.log` were for the page `/login`? (A line looks like `10.0.1.10 - - [12/Oct/2026:08:00:00 +0300] "GET /about.html HTTP/1.1" 404 442`.)

E3. `data/students.csv` has the header `id,name,department,midterm,final`. Print the name and final grade (`name,final`) of the three students with the highest final grades, best first.

Part F --- Write a script (9 points)

Write `passfail.sh`. It takes exactly one argument, a threshold. Without exactly one argument it prints a usage message to standard error and exits with status `1`. Otherwise it counts how many students in `data/students.csv` have a final grade greater than or equal to the threshold ("passed") and how many have less ("failed"), and prints for example `threshold 50: 30 passed, 10 failed`.

— End of the paper. Check your answers. Good luck! —